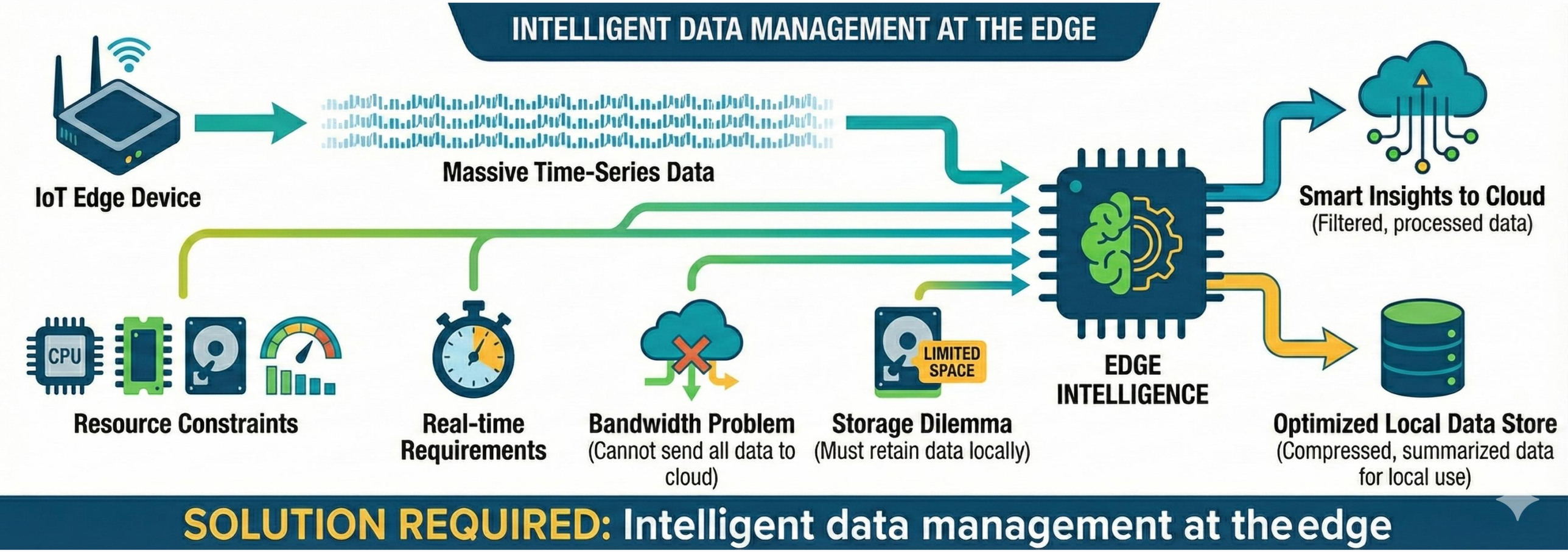
VergeDB

A Task-Aware Time-Series Database for Edge Analytics

Project Need

- IoT Edge Devices: Generate massive time-series data continuously
- Resource Constraints: Limited storage, CPU, memory, bandwidth
- Real-time Requirements: Need immediate local analytics
- Bandwidth Problem: Cannot send all data to cloud
- Storage Dilemma: Must retain data locally but space is limited
- Solution Required: Intelligent data management at the edge

Project Need



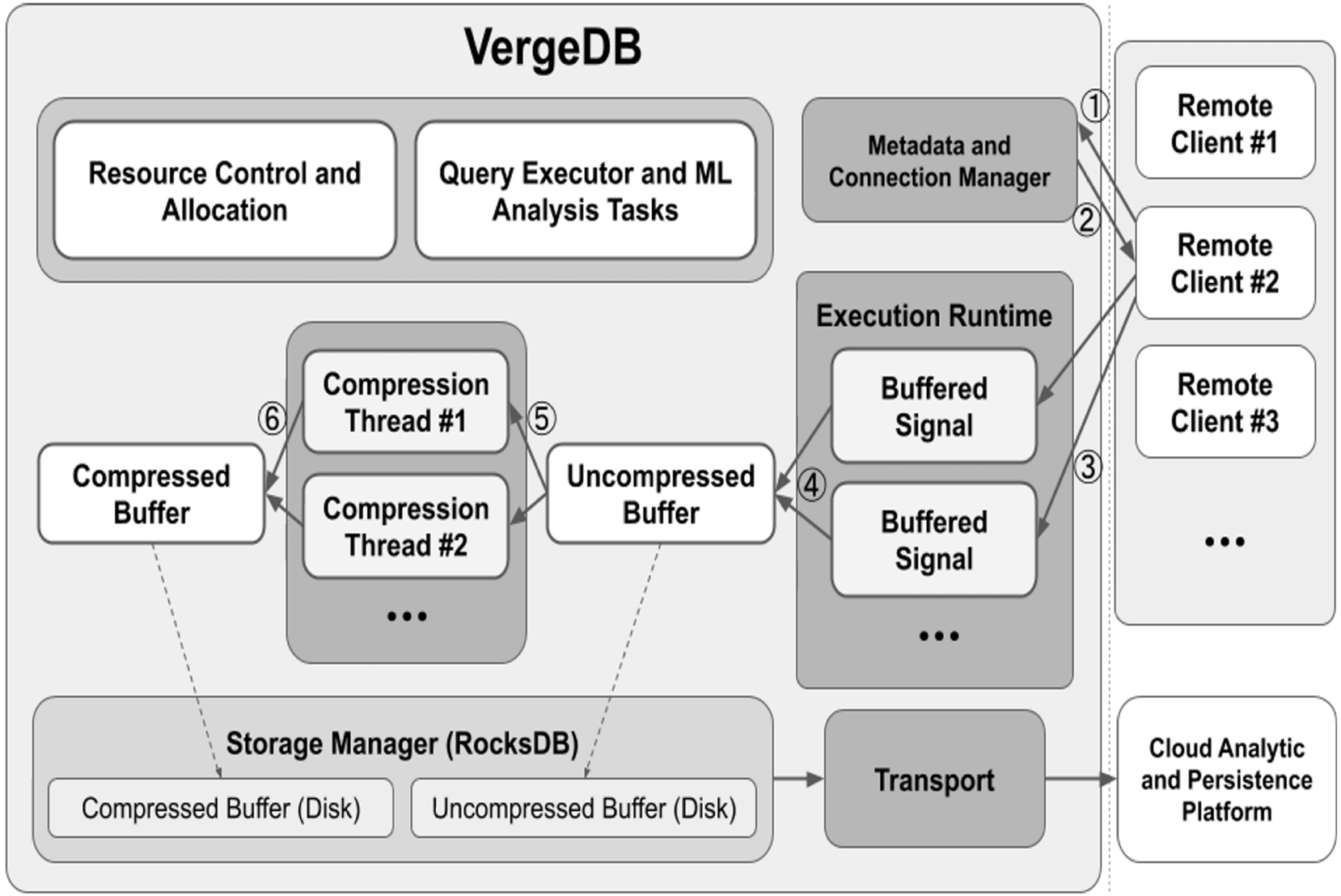
Literature Review Summary

- Paper 1: VergeDB (CIDR 2021) - Task-aware compression
- Paper 2: Edge Transactions - Low latency serializable transactions
- Paper 3: Computing Continuum - IoT to cloud architecture
- Need: Lightweight, adaptive database for edge devices

Why We Selected VergeDB

- Most relevant to edge IoT data management challenge
- Novel task-aware compression approach
- Complete system design with clear architecture
- Addresses real-world edge constraints
- Implementable with measurable performance gains

VergeDB Architecture Diagram



VergeDB Architecture Diagram

Multi-Signal Manager

Handles concurrent data streams from multiple sensors

Compression Engine

8 algorithms: Gzip, Gorilla, Sprintz, PAA, Fourier, etc.

Adaptive Controller

Selects compression based on task + resources

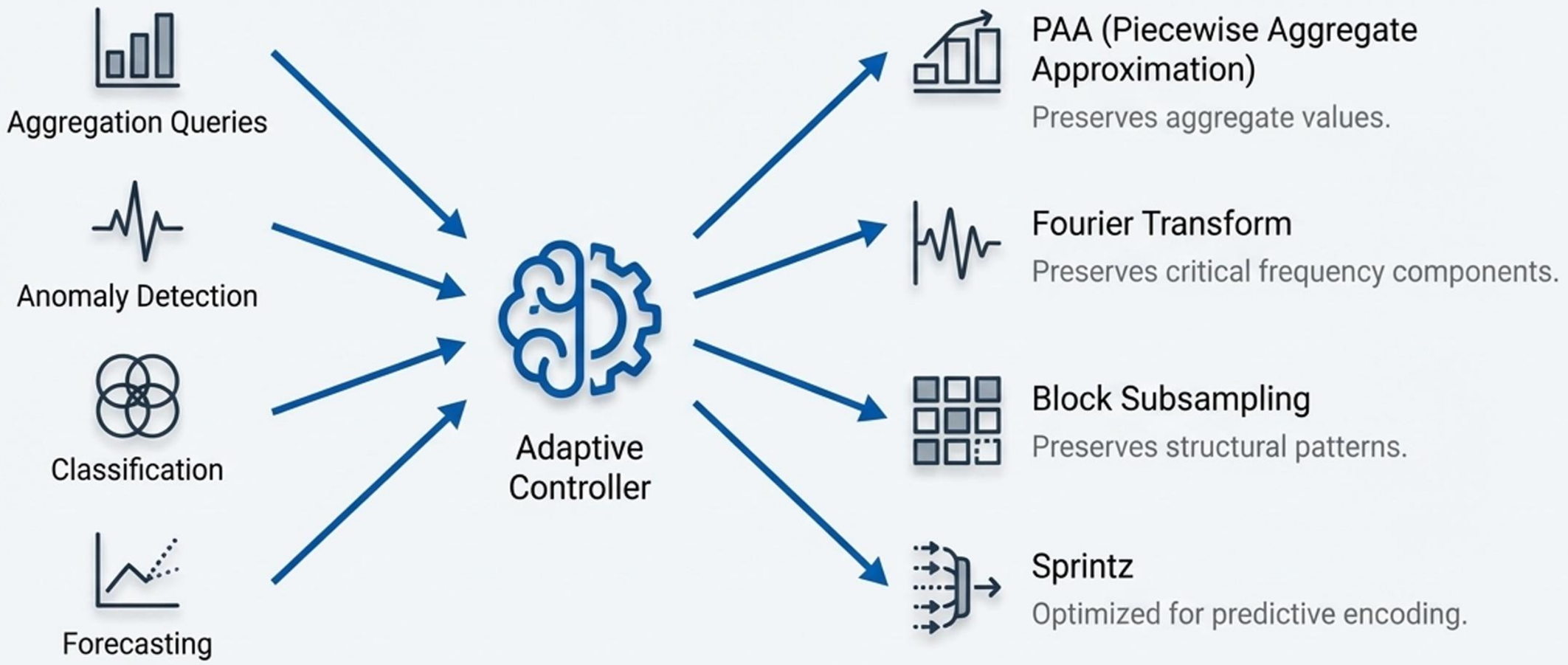
Query Engine

Direct queries on compressed data without decompression

Core Innovation:

Task-Aware Adaptive Compression

Task-Aware Compression Explained



Implementation Components

- verge_database.py: Core engine with multi-signal support
- advanced_compression.py: 8 compression algorithms
- adaptive_controller.py: Resource & task-aware selection
- query_engine.py: Query operations on compressed data
- vergedb_reporter.py: Performance visualization

Performance Results

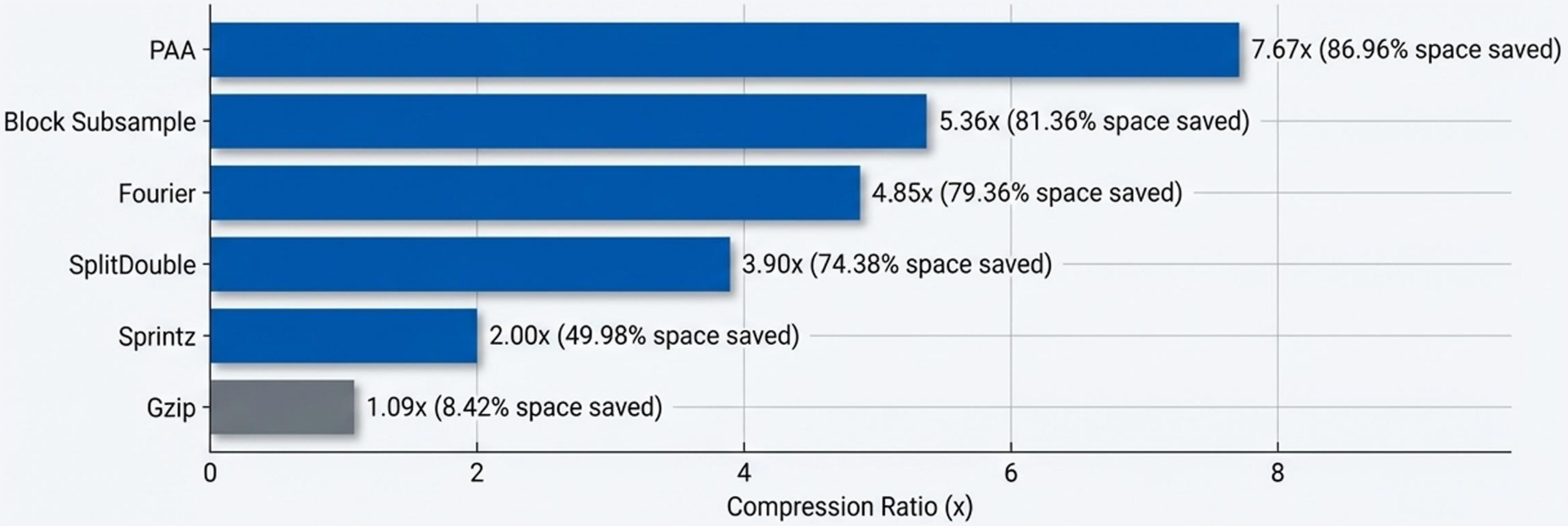
Performance benchmarks based on implementation handling 4 signals and 8,000 data points.

4.68x
Overall Compression Ratio

78.64%
Total Storage Space Saved

2,638 pts/sec
Data Ingestion Rate

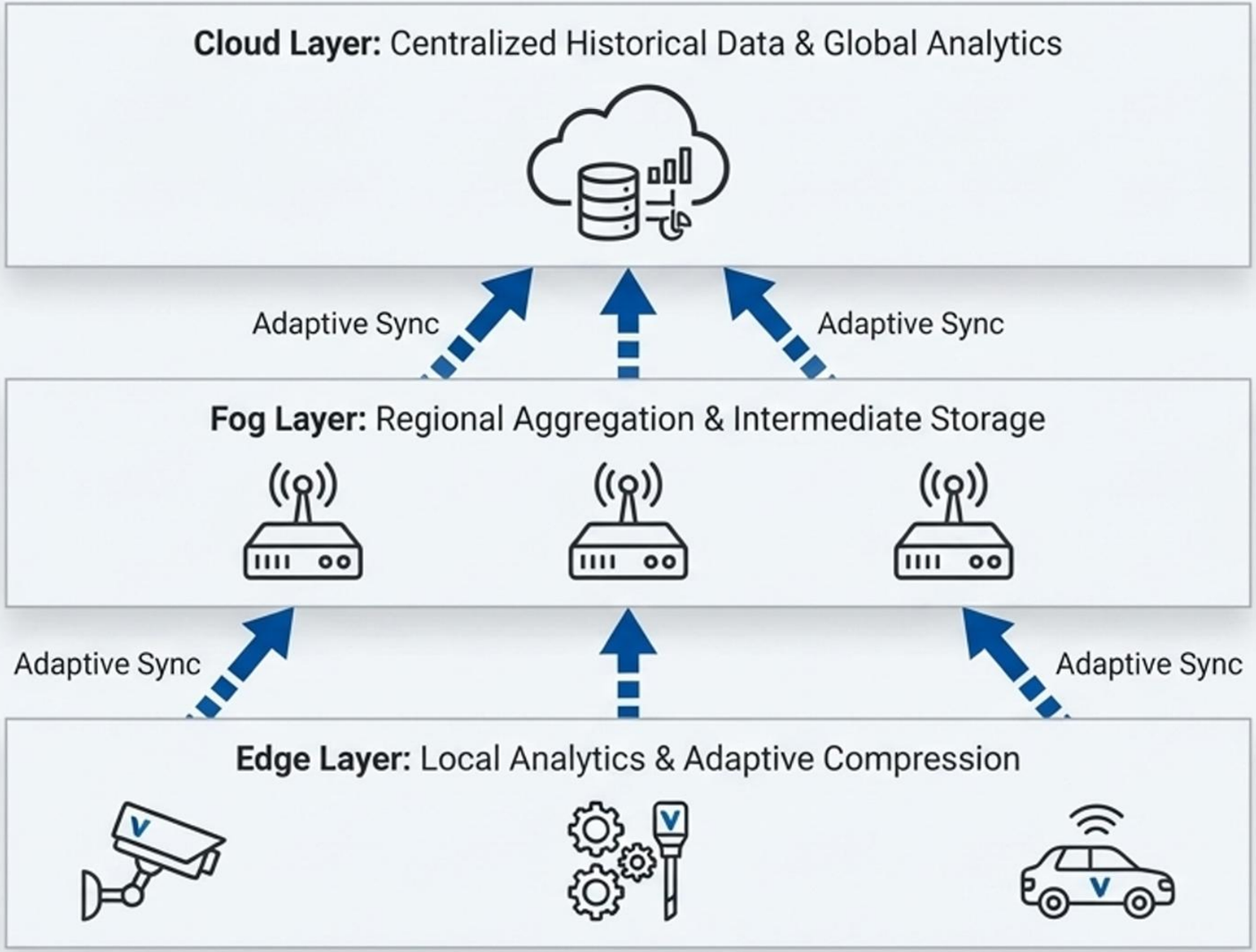
Compression Performance by Method



Result Comparison with Traditional TSDBs

- InfluxDB: Server-focused, high resource usage, no task-aware compression
- Prometheus: Cloud-native, pull-based, not suitable for edge
- TimescaleDB: PostgreSQL overhead, complex setup for edge
- VergeDB Advantages: 78% space savings, query-on-compressed (10-100x faster)
- Edge-First Design: Minimal dependencies, lightweight footprint

Proposed Framework



Key Achievements

- Complete implementation of VergeDB paper (CIDR 2021)
- 8 compression algorithms with task-aware selection
- Query-on-compressed data capability
- 4.68x compression ratio, 78.64% space savings
- Production-ready with comprehensive testing

Future Enhancements

- ML-based compression method selection
- Distributed edge device synchronization
- Advanced query capabilities (joins, windowing)
- GPU-accelerated compression
- Integration with existing IoT platforms

Conclusion

- VergeDB solves critical edge data management challenges
- Task-aware compression: 4.68x ratio, 78.64% savings
- Production-ready implementation with real performance
- Proposed framework enables scalable edge-to-cloud architecture
- Ready for deployment in real-world IoT applications

Thank You!

Questions?